Re-analysis Summary Report for Peer-evaluation

Paper ID: **Barreca_JournPoliEco_2016_J999**

Re-analyst's ID: **z3gmf**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
there was a … decline in the temperature-mortality relationship across decades (p. 133.)

The corresponding article

Here you can find the original article: https://osf.io/fcwj4/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/gpz8h/?view_only=596b97e1bbbc4463ab0887eb399fa019

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
First, I computed a few control variables as done in the original paper: the state-by-month, year-month, as well as the quadratic time variables. Second, I added "decade" as a categorical variable with eleven levels in order to check whether the temperature-mortality relationship declines across all decades. Following the analyses from the paper, a "decade_two" variable was added so that the years before 1959 were categorized as 1 and later years were 2. By comparing the two categories, one will be able to decide whether the temperature-mortality relationship changes from before to after the year 1959. After this, the rows with missing mortality rate data were removed. Additionally, following the paper, I also removed the data from years before 1931 to make the results comparable to what is shown in Table 3. The first regression model (M1) was to test whether there is an association between mortality rate and different days within the ten temperature bins (also using 60-69 $^\circ$F as the baseline). The control variables, same as the original paper, include shares of state population (four age categories) and there interactions with the month indicator, log real per capita income and its interaction with the month indicators, state-by-month, year-by-month, quadratic time effect as well as unusually high or low amounts of precipitation. M1 suggested that days within the nine temperature bins, as compared to the 60-69 $^\circ$F one, had significant positive impact on the mortality rate. Compared with M1, M2 added an interaction with the "decade_two" variable. The results

showed that there were significant interaction between days with average temperature higher than 90 ˚F and the "decade_two" variable. Visualizing this effect with the plot, one can conclude that for days with average temperature higher than 90 ˚F, its association with mortality decreased in the years later than 1959 as compared to the years earlier. However, there is no evidence for this decreased temperature-mortality rate for temperature between 80 and 90 ˚F, partly contradicting what is shown in Table 3. Since the statement made by the paper is regarding "the temperature-mortality relationship across decades", I further tested that whether this association would hold with "decade" as a seven-level categorical variable (from the year 1931 to 2004). As shown with M3 along with the figures, there is some evidence that the temperature-mortality relationship tends to decrease with respect to the days with average temperature higher than 90 ˚F; however, this decrease is not always consistent from decade to decade. M4 and M5 tested more parsimonious models with only the "extreme" temperature bins: lower than 40 ˚F, 80-90 ˚F and higher than 90 ˚F. M4 compared the years before and after 1959, while M5 tested the relationship from decade to decade. Consistent with the previous models, the temperature-mortality rate indeed decreases with respect to the days with average temperature higher than 90 ˚F, but not for days with temperature between 80 and 90 ˚F. Additionally, there was evidence for the decrease in temperature-mortality relationship for days with temperature below 40 ˚F from the years before 1959 to the years after (M4); however, this decrease did not happen from decade to decade.

The analyst's conclusion: There was a decline in the temperature-mortality relationship with respect to the days with more extreme temperature (below 40 ˚F or above 90 ˚ F), when comparing the years before and those after 1959.

The analyst's categorization of the result:
The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should use 90 °F as a threshold for temperature extremes. You should compute the decline between the 1931–59 and 1960–2004 periods. You should conduct your analysis without geographical, or precipitation frequency specification of the data. You should use a 2 months temperature exposure window in your analysis. You should disregard socio-economic factors and technological and public health advancements over time in your analysis. You should not specify the causes of death in your analysis.

Here, the analyst reported the following steps and results of the analysis:
First, I computed a few control variables as done in the original paper: the state-by-month, year-month, as well as the quadratic time variables. Following the instruction of task 2, a "decade_two" variable was added so that the years before 1959 were categorised as 1 and later years were 2. By comparing the two categories, one will be able to decide whether the temperature-mortality relationship changes from before to after the year 1959. After this, the rows with missing mortality rate data were removed. Additionally, also following the instruction of task 2, I removed the data from years before 1931. The regression model (M1) was to test whether there was a significant interaction between the "decade_two" variable and the temperature extremes, which means temperatures that were higher than 90 °F according to the instruction of task 2. The control variables included shares of state population (four age categories) and their interactions with the month indicator, year-by-month, and quadratic time effect. The socioeconomic, geographical and precipitation variables were not included according to the task instruction. The results showed that there was significant interaction between days with average temperature higher than 90 ˚F and the "decade_two" variable ($t = -13.95$, $p<0.001$). Visualizing this effect with the plot, one can conclude that for days with average temperature higher than 90 ˚F, its association with mortality decreased in the years later than 1959 as compared to the years earlier.

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/gzd2m/